Sichi Yan



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Third-year Master's student | Male | 22

Interests: Robot reinforcement-learning locomotion control, humanoid loco-manipulation

Education

Shanghai Jiao Tong University — M.S. in Mechanical Engineering, School of Mechanical Engineering, Robotics Institute · Full-time · Sep 2024 – Jun 2027 (expected)

GPA 3.79/4.0 · Rank 26/274

Shanghai Jiao Tong University — B.E. in Mechanical Engineering, School of Mechanical Engineering · Full-time · Sep 2020 – Jul 2024

Research & Project Experience

Morphology–Control Co-Design for Legged Locomotion in Complex Terrain (Master's research, IEEE RCAR 2026)

Goal: Enable legged robots to adapt to disaster-rescue terrains where a single fixed morphology cannot cover diverse geometric demands.

Approach: Proposed a hierarchical **bi-level co-optimization framework** — outer **CMA-ES** search over continuous morphology parameters, inner rapid **PPO fine-tuning** via reinforcement-learning locomotion priors; a unified parametric morphology model avoids per-candidate asset regeneration.

Outcome: First author of **CALA: Co-design with Accelerated Learning for Terrain-Specific Adaptive Locomotion**, accepted by IEEE RCAR 2026. Validated on **Unitree Go2** and **Ant** across six terrain–platform settings; morphology-randomized pretraining outperforms fixed-morphology and from-scratch baselines, with stronger traversal distance and adaptability on composite sequential terrains.

Learning Crowd Navigation from Human Videos (IEEE ICRA 2026)

Goal: Reduce robot data-collection cost by leveraging easily obtainable human-video demonstrations for safe and efficient robot **navigation** in dynamic crowds.

Approach: Convert UAV-view human videos into BEV occupancy and velocity fields; learn crowd-navigation distributions in SE(2); at test time, project generated trajectories onto each robot platform's kinematic/dynamic feasible set via feasibility guidance — no per-platform retraining. Responsible for **data collection** and **real-robot deployment**.

Outcome: Co-author of **Human2Nav: Learning Crowd Navigation from Human Videos across Robots via Feasibility-Guided Flow Matching**, accepted by ICRA 2026. Validated across four crowd scenarios on real robot platforms; outperforms ORCA, SARL and other baselines in success rate and data efficiency, with cross-platform deployment support.

Lingxi X2 Multi-Humanoid Collaborative Loco-Manipulation (AGIbot, Motion Intelligence R&D Intern, Jun 2026 – present)

Work: Train **multi-humanoid loco-manipulation policies** with PPO-updated Transformers for collaborative carrying and other tasks, with real-robot deployment.

Outcome: Built the full pipeline based on **MuJoCo-Lab**; completed simulation reward design and sim2sim transfer; sim2real in progress.

ROS2 VLM Inference System (Viwistar, Robotics Algorithm Intern, Jun 2025 – Aug 2026)

Goal: Provide near-real-time perception for laboratory automation — recognizing test-tube states and other task anomalies to support downstream robotic manipulation.

Work: Built an end-to-end pipeline from **image collection**, **data annotation**, **LoRA fine-tuning** of Qwen2.5-VL, to **ROS2 service-node deployment**; packaged the model as a trigger-based inference service that captures camera frames, runs inference, and returns compact results on demand.

Outcome: Delivered custom dataset generation, model adaptation, offline validation, and online ROS2 inference deployment; the system returns state judgments (e.g., test tube lying / tilted), providing near-real-time scene understanding for manipulation workflows.

Design and Development of a Handle for Simulating Tricuspid Valve Intervention (Bachelor's Thesis)

Designed a surgical-instrument handle teach pendant and its internal sensing circuitry with **STM32** as the main controller; **SolidWorks** CAD modeling and 3D-printed prototypes; demonstrated with a custom Unity simulator and a cable-driven surgical robot model. Co-inventor of the patent *Tricuspid Valve Interventional Surgery Simulation Data-Acquisition Handle* (Publication No. CN120580913A).

Mutual-Suction Robot for Large Surface Part Machining (Undergraduate Research Project)

Designed **magnetic locking joints for a 6-DOF parallel robot**; ran mechanical and magnetic-field simulations in MATLAB and ANSYS; built prototypes in SolidWorks; completed magnetic-module design and debugging.

Technical Skills

Development tools: Proficient in Python, C++, Linux (Ubuntu), MATLAB, Git, Docker.

Robot control systems: Proficient in Isaac Lab / Isaac Sim and Legged Gym reinforcement-learning training platforms, and the ROS2 communication framework; familiar with multiple RL algorithm frameworks for legged-robot locomotion.

Robot hardware: Proficient in SolidWorks CAD design and modeling, STM32 embedded development; hands-on sim-to-real experience with the Unitree Go2 robot.

Perception & data: Familiar with common sensor hardware (RealSense, Mid-360, etc.) and robot data-collection and processing pipelines.